

Business Banking AI Data Models - Comprehensive Specification

Executive Summary

This specification defines the complete data model architecture for OakNorth's Business Banking AI platform. The design optimizes for natural language querying while maintaining data integrity and supporting complex multi-entity relationships.

Model Architecture Overview

Core Entity Models (7)

1. **mart_businesses** - Legal entities with banking relationships
2. **mart_persons** - Individual humans across all contexts
3. **mart_applications** - Banking application journey records
4. **mart_accounts** - Financial account instances (Spend/Earn)
5. **mart_transactions** - Monetary movements and payments
6. **mart_cards** - Physical payment cards
7. **mart_invitations** - Multi-user business invitations

Bridge Tables (2)

8. **bridge_persons_businesses** - Person-to-business relationships
9. **bridge_persons_cards** - Person-to-card authorizations

Metrics Tables (5)

10. **metrics_businesses_end_of_day** - Daily business aggregations
11. **metrics_accounts_end_of_day** - Daily account metrics
12. **metrics_persons_end_of_day** - Daily person activity
13. **metrics_transactions_end_of_day** - Daily transaction and card metrics
14. **mart_account_balances_end_of_day** - Point-in-time balance snapshots

Supporting Tables (5)

15. **mart_calendar** - Date dimension
16. **mart_sessions** - User activity sessions
17. **mart_interest_accruals** - Daily interest calculations
18. **mart_interest_credits** - Interest payments to customers
19. **metrics_aggregates_end_of_day** - System-wide daily metrics

Communication Tables (2)

20. **mart_emails** - Email campaign and transactional emails

21. **mart_push_alerts** - Mobile push notifications

Detailed Model Specifications

1. **mart_businesses**

Purpose: Single source of truth for all businesses (legal entities) with banking relationships

Grain: One row per business (current state only)

Field Name	Data Type	Description
business_id	STRING	Primary key - unique business identifier
companies_house_number	STRING	8-digit Companies House number
business_name	STRING	Legal entity name from Companies House
trading_name	STRING	Trading name if different from business name
incorporated_on	DATE	Date of company incorporation
entity_type	STRING	Legal structure (Ltd, plc, LLP, etc.)
entity_status	STRING	Companies House status (Active, dissolved, etc.)
sic_code	STRING	Primary Standard Industrial Classification code
sic_description	STRING	Industry description
secondary_sic_codes	STRING	Comma-separated additional SIC codes
registered_office_address_line_1	STRING	Registered address line 1
registered_office_address_line_2	STRING	Registered address line 2
registered_office_address_care_of	STRING	Care of name
registered_office_address_po_box	STRING	PO box number
registered_office_address_premises	STRING	Building name/number
registered_office_address_locality	STRING	Town/city
registered_office_address_region	STRING	County/region
registered_office_address_country	STRING	Country
registered_office_address_postal_code	STRING	Postcode (formatted with space)
correspondence_address_line_1	STRING	Mailing address if different from registered office
correspondence_address_line_2	STRING	Mailing address line 2
correspondence_address_postal_code	STRING	Mailing postcode
annual_turnover	STRING	Turnover band from applicable accounts
headcount	STRING	Employee count band
annual_balance_sheet_total	NUMERIC	Balance sheet total (£)
is_micro_enterprise	BOOLEAN	Consumer duty classification
is_micro_small_medium_enterprise	BOOLEAN	FSCS classification
is_fscs_eligible	BOOLEAN	FSCS protection eligibility
fscs_notes	STRING	FSCS eligibility explanation
fscs_depositor_accounts_status_codes	STRING	Account status codes (/) separated

Field Name	Data Type	Description
is_fscs_depositor_sanctioned	BOOLEAN	Sanctions exclusion flag
are_fscs_depositor_accounts_legally_disputed	BOOLEAN	Legal dispute exclusion
is_fscs_depositor_legally_dormant	BOOLEAN	Dormancy exclusion
are_fscs_depositor_accounts_beneficiary	BOOLEAN	Beneficiary exclusion
cdd_risk_rating	STRING	Customer due diligence rating
cdd_risk_score	NUMERIC	Numeric risk score
due_diligence_type	STRING	Type of DD performed
is_fatca_crs_reportable	BOOLEAN	Tax reporting requirement
do_company_or_controlling_persons_hold_outside_uk_tax_residency	BOOLEAN	Tax residency flag
do_controlling_persons_hold_us_citizenship	BOOLEAN	US person flag
source_of_funds_sources	STRING	Comma-separated funding sources
account_purpose	STRING	Intended use of accounts
lifecycle_stage	STRING	Prospect/Applicant/Customer Customer
service_status	STRING	Active/Suspended/Blocked
application_started_at	TIMESTAMP	First application timestamp
application_approved_at	TIMESTAMP	CDD approval timestamp
activated_at	TIMESTAMP	First transaction timestamp
suspended_at	TIMESTAMP	Service suspension timestamp
removed_at	TIMESTAMP	Relationship termination timestamp
is_active	BOOLEAN	Currently operational flag
is_activated	BOOLEAN	Has completed first transaction
has_open_accounts	BOOLEAN	Has active financial accounts
has_active_cards	BOOLEAN	Has operational payment cards
has_accounting_integration	BOOLEAN	Connected to accounting system
accounting_integration_total_synced_credit_amount	NUMERIC	Total credits synced
accounting_integration_total_synced_debit_amount	NUMERIC	Total debits synced
accounting_integration_total_synced_transaction_count	NUMERIC	Total transactions synced
is_multi_entity_business	BOOLEAN	Multiple persons associated
is_internal_test	BOOLEAN	Internal testing account
compass_link	STRING	Internal portal URL
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Last modification timestamp

2. mart_persons

Purpose: Core identity for all individuals, independent of business relationships

Grain: One row per person (current state only)

Field Name	Data Type	Description
person_id	STRING	Primary key - unique person identifier (maps to user_id)
auth0_id	STRING	Authentication system identifier
full_name	STRING	Complete legal name
first_name	STRING	Given name
last_name	STRING	Surname
middle_names	STRING	Middle names if provided
email_address	STRING	Primary email
phone_number	STRING	UK formatted mobile (07XXXXXXXXXX)
date_of_birth	DATE	For age verification
nationality	STRING	Primary nationality
country_of_residence	STRING	Current residence country
primary_address_line_1	STRING	Home address line 1
primary_address_line_2	STRING	Home address line 2
primary_address_line_3	STRING	Home address line 3
primary_address_line_4	STRING	Home address line 4
primary_address_city_town	STRING	City/town
primary_address_county	STRING	County
primary_address_post_code	STRING	Postcode (formatted)
primary_address_country_code	STRING	Country code
primary_address_care_of	STRING	Care of field
idv_status	STRING	Identity verification status
idv_provider	STRING	IDV service used (MiTek, etc.)
idv_completed_at	TIMESTAMP	IDV completion timestamp
idv_document_type	STRING	ID document used
idv_document_full_name	STRING	Name on ID document
idv_document_date_of_birth	DATE	DOB on ID document
is_identity_verified	BOOLEAN	Passed IDV check
is_pep	BOOLEAN	Politically exposed person
is_sanctioned	BOOLEAN	On sanctions list
overall_lifecycle_stage	STRING	Highest stage across businesses
overall_service_status	STRING	Worst status across businesses
is_multi_entity_person	BOOLEAN	Associated with multiple businesses
businesses_count	NUMERIC	Number of business associations
active_businesses_count	NUMERIC	Number of active relationships
first_application_at	TIMESTAMP	Earliest application

Field Name	Data Type	Description
first_activated_at	TIMESTAMP	First became User
last_login_at	TIMESTAMP	Most recent authentication
last_activity_at	TIMESTAMP	Most recent action
total_sessions_count	NUMERIC	Lifetime login count
personal_needs	STRING	Accessibility requirements (/ separated)
origin	STRING	How person initially joined
origin_business_application_id	STRING	Initial business application
is_internal_user	BOOLEAN	OakNorth employee flag
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Last modification timestamp

3. mart_applications

Purpose: Complete application journey from submission to decision

Grain: One row per application

Field Name	Data Type	Description
application_id	STRING	Primary key - unique application identifier
person_id	STRING	Applicant person ID
business_id	STRING	Associated business (postcode)
companies_house_number	STRING	Target company number
application_flow_type	STRING	cold_start/multi_company
application_channel	STRING	mobile_app/web_app
application_version	STRING	Form version used
started_at	TIMESTAMP	Application initiation
submitted_at	TIMESTAMP	Full submission timestamp
idv_started_at	TIMESTAMP	Identity verification start
idv_completed_at	TIMESTAMP	Identity verification complete
idv_attempts_count	NUMERIC	Number of IDV attempts
idv_outcome	STRING	approved/retake
cdd_assigned_at	TIMESTAMP	Assigned to reviewer
cdd_started_at	TIMESTAMP	Review commenced
cdd_decision_at	TIMESTAMP	Decision timestamp
cdd_decision_status	STRING	APPROVED/REJECTED/WITHDRAWN
cdd_reviewer_id	STRING	Reviewing officer ID
cdd_reviewer_name	STRING	Reviewer full name
rejection_reason	STRING	If rejected, primary reason
rejection_subcategory	STRING	Detailed rejection category
business_name	STRING	Company name applied for
trading_name	STRING	If different from legal name
website_url	STRING	Company website
business_address_line_1	STRING	Trading address line 1
business_address_line_2	STRING	Trading address line 2
business_address_postcode	STRING	Trading postcode
is_correspondence_address_different	BOOLEAN	Different from registered
account_purpose	STRING	Primary intended use
secondary_purposes	STRING	Additional use cases
estimated_initial_deposit	STRING	Deposit range band
estimated_monthly_deposits	STRING	Monthly inflow band
estimated_monthly_withdrawals	STRING	Monthly outflow band
estimated_turnover	STRING	Annual revenue band

Field Name	Data Type	Description
headcount	STRING	Employee count band
source_of_funds_sources	STRING	Funding source types
source_of_funds_sale_of_assets_address	STRING	If asset sale
source_of_funds_capital_from_third_party_legal_name	STRING	If third party
source_of_funds_grants_or_subsidies_provider	STRING	If grants
source_of_funds_loan_or_finance_legal_name	STRING	If loan
source_of_funds_other	STRING	Other sources
source_of_funds_rental_income_address	STRING	If rental
existing_banking_provider	STRING	Current bank name
reason_for_switching	STRING	Why leaving current bank
intended_payment_types	STRING	Payment methods needed
international_payments_needed	BOOLEAN	Cross-border requirement
countries_of_operation	STRING	Operating countries
is_regulated_business	BOOLEAN	FCA/other regulation
regulatory_body	STRING	If regulated, which body
has_business_plan	BOOLEAN	Formal plan exists
has_financial_projections	BOOLEAN	Forecasts available
utm_source	STRING	Marketing attribution
utm_medium	STRING	Marketing channel
utm_campaign	STRING	Campaign name
referrer_type	STRING	Referral source type
referrer_name	STRING	Specific referrer
person_email_address	STRING	Applicant email
person_phone_number	STRING	Applicant phone
person_date_of_birth	DATE	Applicant DOB
person_address_postcode	STRING	Applicant postcode
is_micro_small_medium_enterprise	BOOLEAN	FSCS classification
is_micro_enterprise	BOOLEAN	Consumer duty classification
annual_balance_sheet_total	NUMERIC	Financial size
is_fatca_crs_reportable_account	BOOLEAN	Tax reporting
is_fscs_eligible	BOOLEAN	FSCS protection
fscs_notes	STRING	FSCS explanation
fscs_depositor_accounts_status_codes	STRING	Status codes
do_company_or_controlling_persons_hold_outside_uk_tax_residency	BOOLEAN	Tax flag
do_controlling_persons_hold_us_citizenship	BOOLEAN	US person flag

Field Name	Data Type	Description
is_multi_company_application	BOOLEAN	Additional company flag
is_initial_application	BOOLEAN	First time applicant
is_internal_user	BOOLEAN	Internal test account
days_to_submission	NUMERIC	From start to submit
days_to_decision	NUMERIC	From submit to decision
days_from_submission_to_idv	NUMERIC	IDV timing
created_at	TIMESTAMP	Record creation
updated_at	TIMESTAMP	Last modification

4. mart_accounts

Purpose: Individual financial account instances

Grain: One row per account (current state only)

Field Name	Data Type	Description
account_id	STRING	Primary key - unique account identifier (maps to id)
business_id	STRING	Owning business
companies_house_number	STRING	For easier joins
account_number	STRING	8-digit account number
sort_code	STRING	6-digit sort code
iban	STRING	International identifier
bic_swift	STRING	SWIFT code
account_type	STRING	Spend Account/Earn Account
account_subtype	STRING	Detailed product variant
product_name	STRING	Customer-facing name
product_id	STRING	Product identifier
product_version_id	STRING	Product iteration
platform	STRING	thought_machine/mambu
ledger_id	STRING	Core banking reference
currency	STRING	GBP (always currently)
denomination	STRING	Currency denomination
interest_aer	NUMERIC	Annual equivalent rate (deposit_interest_aer)
interest_calculation_method	STRING	How interest computed
interest_payment_frequency	STRING	monthly/quarterly/annually
interest_accrual_schedule_hour	NUMERIC	Hour of day (0-23)
interest_accrual_schedule_minute	NUMERIC	Minute of hour (0-59)
interest_application_schedule_day	NUMERIC	Day of month (1-31)
interest_application_schedule_hour	NUMERIC	Hour of day (0-23)
interest_application_schedule_minute	NUMERIC	Minute of hour (0-59)
next_interest_payment_date	DATE	Upcoming payment
notice_period_days	NUMERIC	Withdrawal notice required
minimum_balance	NUMERIC	If applicable
maximum_balance	NUMERIC	If applicable
maximum_total_daily_outbound_payments_amount	NUMERIC	Daily limit
overdraft_limit	NUMERIC	Arranged overdraft
monthly_fee	NUMERIC	Account charges
transaction_fees	STRING	JSON - fee structure
t_side	STRING	Balance sheet side

Field Name	Data Type	Description
opened_at	TIMESTAMP	Account creation
activated_at	TIMESTAMP	First transaction
first_deposit_date	DATE	Initial funding
first_deposit_amount	NUMERIC	Initial funding value
last_transaction_date	DATE	Most recent activity
last_transaction_type	STRING	Type of last transaction
suspended_at	TIMESTAMP	If suspended
suspension_reason	STRING	Why suspended
closed_at	TIMESTAMP	Closure timestamp (removed_at)
closure_reason	STRING	Closure reason
is_active	BOOLEAN	Currently operational
is_primary_account	BOOLEAN	Main business account
has_overdraft	BOOLEAN	Overdraft facility exists
has_pending_transactions	BOOLEAN	Unsettled transactions
has_accounting_integration	BOOLEAN	Connected to accounting
accounting_integration_last_sync_status	STRING	Sync status
accounting_integration_last_sync_at	TIMESTAMP	Last sync time
accounting_integration_last_successful_sync_at	TIMESTAMP	Last success
accounting_integration_latest_connection_status	STRING	Connection state
accounting_integration_total_synced_credit_amount	NUMERIC	Credits synced
accounting_integration_total_synced_debit_amount	NUMERIC	Debits synced
accounting_integration_total_synced_transaction_count	NUMERIC	Count synced
accounting_integration_initiator_person_ids	STRING	Who connected
total_deposits_count	NUMERIC	Lifetime inbound count
total_deposits_amount	NUMERIC	Lifetime inbound value
total_withdrawals_count	NUMERIC	Lifetime outbound count
total_withdrawals_amount	NUMERIC	Lifetime outbound value
current_balance	NUMERIC	Latest settled balance
available_balance	NUMERIC	Including overdraft
pending_balance	NUMERIC	Unsettled amount
days_since_last_transaction	NUMERIC	Inactivity measure
is_dormant	BOOLEAN	No recent activity
is_internal_user	BOOLEAN	Internal test account
compass_link	STRING	Internal portal URL
created_at	TIMESTAMP	Record creation

Field Name	Data Type	Description
updated_at	TIMESTAMP	Last modification

5. mart_transactions

Purpose: All monetary movements - customer-facing view (combines customer_ledger_transactions insights)

Grain: One row per transaction per account (transactions touching two accounts appear twice)

Field Name	Data Type	Description
transaction_id	STRING	Primary key component - ledger transaction ID
padded_transaction_id	STRING	36-char version with 'a' prefix
account_id	STRING	Primary key component - associated account
business_id	STRING	For easier analysis
companies_house_number	STRING	For easier joins
transaction_reference	STRING	Customer reference
payment_reference	STRING	Bank reference (reference field)
direction	STRING	inbound/outbound
type	STRING	fps/bacs/chaps/card/internal transfer/interest payment
method	STRING	Payment method used
channel	STRING	mobile_app/web_app/system
amount	NUMERIC	Transaction value (settled or instructed)
currency	STRING	Transaction currency
denomination	STRING	Currency denomination
exchange_rate	NUMERIC	If non-GBP
fee_amount	NUMERIC	Associated charges
vat_amount	NUMERIC	VAT on fees
status	STRING	pending/completed/cancelled
status_reason	STRING	If not completed
counterparty_name	STRING	Other party name
counterparty_account_number	STRING	Other party account
counterparty_sort_code	STRING	Other party bank
counterparty_iban	STRING	International identifier
counterparty_reference	STRING	Their reference
confirmation_of_payee_match_type	STRING	exact/close/no match/null
confirmation_of_payee_match_code	STRING	Detailed CoP code
confirmation_of_payee_response	STRING	Full CoP response
scheme_reference	STRING	Payment scheme ID
settlement_date	DATE	When funds settled
value_date	DATE	Interest calculation date
created_at	TIMESTAMP	Initiation time
completed_at	TIMESTAMP	Settlement time
failed_at	TIMESTAMP	If failed
cancelled_at	TIMESTAMP	If cancelled
last_updated_at	TIMESTAMP	Last modification

Field Name	Data Type	Description
created_by_person_id	STRING	Who initiated
created_by_person_name	STRING	Initiator name
approved_by_person_id	STRING	If approval required
approved_at	TIMESTAMP	Approval timestamp
is_scheduled	BOOLEAN	Future-dated payment
scheduled_for	DATE	Intended date
is_recurring	BOOLEAN	Standing order
recurrence_pattern	STRING	If recurring
recurrence_end_date	DATE	Last occurrence
is_international	BOOLEAN	Cross-border
beneficiary_bank_country	STRING	Destination country
intermediary_bank	STRING	Correspondent bank
purpose_code	STRING	International purpose
is_first_transaction	BOOLEAN	Triggers activation
is_salary_payment	BOOLEAN	Payroll identified
is_tax_payment	BOOLEAN	HMRC payment
is_card_transaction	BOOLEAN	Card-originated
is_external_payment	BOOLEAN	Not internal transfer
merchant_category_code	STRING	If card transaction
merchant_name	STRING	Card acceptor
card_id	STRING	If card transaction
risk_score	NUMERIC	Transaction risk rating
fraud_check_result	STRING	Screening outcome
sanctions_check_result	STRING	Sanctions screening
is_held_for_review	BOOLEAN	Manual review required
review_completed_at	TIMESTAMP	Review completion
reviewer_decision	STRING	Review outcome
batch_id	STRING	If part of batch
batch_position	NUMERIC	Order in batch
original_transaction_id	STRING	If refund/reversal
is_reversal	BOOLEAN	Reversal flag
is_refund	BOOLEAN	Refund flag
metadata	STRING	JSON - additional data
is_internal_user	BOOLEAN	Internal test transaction
created_at_record	TIMESTAMP	Record creation

Field Name	Data Type	Description
updated_at_record	TIMESTAMP	Record modification

6. mart_cards

Purpose: Physical payment cards lifecycle and configuration

Grain: One row per card (current state only)

Field Name	Data Type	Description
card_id	STRING	Primary key - unique card identifier
person_id	STRING	Primary cardholder (cardholder_user_id)
account_id	STRING	Linked account (ledger_account_id)
business_id	STRING	For easier analysis
companies_house_number	STRING	For easier joins
card_number_masked	STRING	First 6, last 4 digits (partial_card_number)
card_type	STRING	debit/credit/prepaid
card_brand	STRING	visa/mastercard
card_design	STRING	Physical design variant
cardholder_name	STRING	Name on card
is_primary_card	BOOLEAN	Main card for account
requested_at	TIMESTAMP	Order placement
requested_by_person_id	STRING	Who ordered
approved_at	TIMESTAMP	Order approval
approved_by_person_id	STRING	Who approved
manufactured_at	TIMESTAMP	Production date
shipped_at	TIMESTAMP	Dispatch date
delivered_at	TIMESTAMP	Delivery confirmation
activated_at	TIMESTAMP	First use (card_activated_at)
activation_method	STRING	app/atm/online
expires_at	TIMESTAMP	Expiry date (card_expiring_at)
four_digit_expiration_date	STRING	MM/YY format
pin_set_at	TIMESTAMP	PIN creation
has_pin_set	BOOLEAN	PIN configured
pin_attempts_count	NUMERIC	Failed PIN entries
last_pin_attempt_at	TIMESTAMP	Recent PIN try
state	STRING	active/suspended/terminated/unactivated/limited
fulfillment_status	STRING	ordered/shipped/delivered/digitally_presented
reason_code	STRING	State change reason code
reason_description	STRING	State change description
event_type	STRING	Latest event type
suspended_at	TIMESTAMP	If suspended
suspended_by_person_id	STRING	Who suspended
suspension_reason	STRING	Why suspended
terminated_at	TIMESTAMP	If terminated

Field Name	Data Type	Description
terminated_by_person_id	STRING	Who terminated
termination_reason	STRING	Why terminated
replacement_card_id	STRING	If replaced
replaces_card_id	STRING	If replacement
daily_atm_limit	NUMERIC	Cash withdrawal limit
daily_purchase_limit	NUMERIC	Spending limit
daily_online_limit	NUMERIC	E-commerce limit
international_enabled	BOOLEAN	Cross-border usage
contactless_enabled	BOOLEAN	Tap to pay
atm_enabled	BOOLEAN	Cash withdrawals
online_enabled	BOOLEAN	E-commerce
gambling_blocked	BOOLEAN	Merchant restriction
adult_blocked	BOOLEAN	Merchant restriction
last_transaction_at	TIMESTAMP	Most recent use
last_transaction_amount	NUMERIC	Recent transaction
last_transaction_merchant	STRING	Where used
last_transaction_country	STRING	Usage location
total_transactions_count	NUMERIC	Lifetime usage
total_transactions_amount	NUMERIC	Lifetime spend
total_atm_count	NUMERIC	Cash withdrawals
total_atm_amount	NUMERIC	Cash withdrawn
fraud_alerts_count	NUMERIC	Suspicious activity
is_active	BOOLEAN	Currently usable
is_virtual	BOOLEAN	Digital-only card
is_internal_user	BOOLEAN	Internal test card
created_at	TIMESTAMP	Record creation (card_created_at)
updated_at	TIMESTAMP	Last modification

7. mart_invitations

Purpose: Track multi-user business invitation lifecycle

Grain: One row per invitation

Field Name	Data Type	Description
invitation_id	STRING	Primary key - unique invitation identifier
business_id	STRING	Target business
business_application_id	STRING	Associated application
companies_house_number	STRING	For easier joins
invitation_type	STRING	new_user/existing_user
invitation_reason	STRING	Why invited
invitee_email	STRING	Recipient email
invitee_full_name	STRING	Recipient name
invitee_phone_number	STRING	Recipient phone
invitee_position_title	STRING	Professional title
invitee_position_type	STRING	director/other
invitee_department	STRING	Business unit
is_existing_person	BOOLEAN	Already in system (invitee_is_existing_person)
existing_person_id	STRING	If already exists (invitee_auth0_user_id)
invitee_payment_role	STRING	Payment permissions
invitee_should_be_admin	BOOLEAN	Admin role intended
intended_permissions	STRING	JSON - detailed permissions
payment_approver	BOOLEAN	Payment authority
payment_approval_limit	NUMERIC	Maximum approval
can_manage_cards	BOOLEAN	Card administration
can_invite_users	BOOLEAN	User management
can_view_all_accounts	BOOLEAN	Full visibility
created_by_person_id	STRING	Inviter ID (created_by_user_id)
created_by_name	STRING	Inviter name (created_by_user_name)
created_by_role	STRING	Inviter's role
created_at	TIMESTAMP	Invitation sent
reminder_sent_at	TIMESTAMP	Follow-up sent
reminder_count	NUMERIC	Follow-ups sent
expires_at	TIMESTAMP	Invitation expiry
claimed_at	TIMESTAMP	Acceptance time
accepted_from_ip	STRING	Acceptance location
accepted_from_device	STRING	Device type
deleted_at	TIMESTAMP	If withdrawn
deleted_by_person_id	STRING	Who deleted
deletion_reason	STRING	Why deleted

Field Name	Data Type	Description
status	STRING	pending/approved/cancelled/expired
origin	STRING	Where invitation originated
invitation_token	STRING	Unique URL token
onboarding_completed_at	TIMESTAMP	Setup finished
first_login_at	TIMESTAMP	Initial access
is_active	BOOLEAN	Currently valid
is_internal_user	BOOLEAN	Internal test invitation
created_at_record	TIMESTAMP	Record creation
updated_at	TIMESTAMP	Last modification

8. bridge_persons_businesses

Purpose: Many-to-many relationships between persons and businesses

Grain: One row per person-business relationship

Field Name	Data Type	Description
person_id	STRING	Person identifier (user_id)
business_id	STRING	Business identifier
companies_house_number	STRING	For easier joins
relationship_type	STRING	applicant/invitee/admin_assigned
role	STRING	admin/member
lifecycle_stage	STRING	invited/applicant/user/ex_user
service_status	STRING	active/suspended
permissions_template	STRING	Standard permission set
payment_approver	BOOLEAN	Can approve payments
payment_approval_limit	NUMERIC	Maximum approval amount
can_manage_cards	BOOLEAN	Card administration
can_invite_users	BOOLEAN	User management
can_view_all_accounts	BOOLEAN	Full account visibility
can_export_data	BOOLEAN	Data export permission
can_manage_integrations	BOOLEAN	Integration management
joined_at	TIMESTAMP	Relationship start
activated_at	TIMESTAMP	First login as user
suspended_at	TIMESTAMP	If suspended
suspension_reason	STRING	Why suspended
removed_at	TIMESTAMP	Relationship end
removal_reason	STRING	Why removed
is_primary_business	BOOLEAN	Main business relationship
is_original_applicant	BOOLEAN	Founded the business account
invitation_id	STRING	Source invitation if applicable
added_by_person_id	STRING	Who added this person
is_active	BOOLEAN	Currently active relationship
created_at	TIMESTAMP	Record creation
updated_at	TIMESTAMP	Last modification

9. bridge_persons_cards

Purpose: Link persons to cards they're authorized to use

Grain: One row per person-card authorization

Field Name	Data Type	Description
person_id	STRING	Person identifier
card_id	STRING	Card identifier
authorization_type	STRING	primary/additional
authorized_at	TIMESTAMP	When authorized
authorized_by_person_id	STRING	Who authorized
daily_limit	NUMERIC	Personal spending limit
monthly_limit	NUMERIC	Monthly spending limit
can_use_internationally	BOOLEAN	International permission
can_use_online	BOOLEAN	E-commerce permission
can_withdraw_cash	BOOLEAN	ATM permission
merchant_category_restrictions	STRING	Blocked categories
revoked_at	TIMESTAMP	If authorization revoked
revoked_by_person_id	STRING	Who revoked
revocation_reason	STRING	Why revoked
is_active	BOOLEAN	Currently authorized
created_at	TIMESTAMP	Record creation
updated_at	TIMESTAMP	Last modification

10. metrics_businesses_end_of_day

Purpose: Daily business-level aggregated metrics

Grain: One row per business per day

Field Name	Data Type	Description
date	DATE	Metrics date
business_id	STRING	Business identifier
companies_house_number	STRING	For easier joins
business_name	STRING	For easier analysis
lifecycle_stage	STRING	As of date
service_status	STRING	As of date
days_since_application	NUMERIC	From first application
days_since_activation	NUMERIC	From first transaction
days_since_last_transaction	NUMERIC	Inactivity measure
total_balance	NUMERIC	Sum across all accounts
total_available_balance	NUMERIC	Including overdrafts
spend_balance	NUMERIC	Current account total
earn_balance	NUMERIC	Savings account total
pending_inbound	NUMERIC	Unsettled credits
pending_outbound	NUMERIC	Unsettled debits
balance_change_from_previous_day	NUMERIC	Daily movement
max_balance_to_date	NUMERIC	Historical peak
is_above_1k_balance	BOOLEAN	Balance threshold
is_above_10k_balance	BOOLEAN	Balance threshold
is_above_100k_balance	BOOLEAN	Balance threshold
days_above_1k_balance	NUMERIC	Cumulative days
days_above_10k_balance	NUMERIC	Cumulative days
days_above_100k_balance	NUMERIC	Cumulative days
transaction_count	NUMERIC	Daily transactions
transaction_volume	NUMERIC	Daily value
inbound_count	NUMERIC	Daily deposits
inbound_volume	NUMERIC	Daily deposit value
outbound_count	NUMERIC	Daily withdrawals
outbound_volume	NUMERIC	Daily withdrawal value
unique_transacting_accounts	NUMERIC	Active accounts
card_transaction_count	NUMERIC	Card usage
card_transaction_volume	NUMERIC	Card spend
international_transaction_count	NUMERIC	Cross-border
active_persons_count	NUMERIC	Logged in today
total_persons_count	NUMERIC	All associated

Field Name	Data Type	Description
new_persons_added	NUMERIC	Joined today
total_cards_count	NUMERIC	All cards
active_cards_count	NUMERIC	Usable cards
new_cards_activated	NUMERIC	Activated today
total_logins	NUMERIC	Session count
unique_login_persons	NUMERIC	Distinct users
mobile_logins	NUMERIC	Mobile sessions
web_logins	NUMERIC	Web sessions
is_active_today	BOOLEAN	Any activity
has_transactions_today	BOOLEAN	Transaction flag
has_logins_today	BOOLEAN	Login flag
created_at	TIMESTAMP	Record creation

11. metrics_accounts_end_of_day

Purpose: Daily account-level metrics

Grain: One row per account per day

Field Name	Data Type	Description
date	DATE	Metrics date
account_id	STRING	Account identifier
business_id	STRING	For easier joins
companies_house_number	STRING	For easier joins
account_type	STRING	Spend/Earn
account_number	STRING	For identification
lifecycle_stage	STRING	As of date
service_status	STRING	As of date
days_since_opened	NUMERIC	From creation
days_since_activated	NUMERIC	From first transaction
days_since_last_transaction	NUMERIC	Inactivity
settled_balance	NUMERIC	End of day balance
pending_inbound	NUMERIC	Unsettled credits
pending_outbound	NUMERIC	Unsettled debits
available_balance	NUMERIC	Including overdraft
minimum_balance	NUMERIC	Daily low
maximum_balance	NUMERIC	Daily high
balance_change	NUMERIC	From previous day
transaction_count	NUMERIC	Daily count
inbound_count	NUMERIC	Deposits
outbound_count	NUMERIC	Withdrawals
largest_inbound	NUMERIC	Biggest deposit
largest_outbound	NUMERIC	Biggest withdrawal
unique_counterparties	NUMERIC	Distinct parties
card_transaction_count	NUMERIC	Card usage
card_transaction_volume	NUMERIC	Card spend
fee_amount	NUMERIC	Daily fees
interest_accrued	NUMERIC	For Earn accounts
interest_rate	NUMERIC	Effective rate
is_active_today	BOOLEAN	Any activity
has_transactions_today	BOOLEAN	Transaction flag
is_overdrawn	BOOLEAN	Negative balance
created_at	TIMESTAMP	Record creation

12. metrics_persons_end_of_day

Purpose: Daily person engagement metrics

Grain: One row per person per day

Field Name	Data Type	Description
date	DATE	Metrics date
person_id	STRING	Person identifier
overall_lifecycle_stage	STRING	As of date
overall_service_status	STRING	As of date
active_businesses_count	NUMERIC	Current associations
sessions_count	NUMERIC	Daily logins
total_session_duration_minutes	NUMERIC	Time in app
pages_viewed	NUMERIC	Screens visited
actions_performed	NUMERIC	Buttons clicked
primary_platform	STRING	web_app/mobile_app
devices_used	NUMERIC	Distinct devices
features_used	STRING	Comma-separated list
transactions_initiated	NUMERIC	Payments started
transactions_completed	NUMERIC	Payments finished
transactions_approved	NUMERIC	Others' payments
total_transaction_value	NUMERIC	Daily volume
cards_used_count	NUMERIC	Distinct cards
card_transactions_count	NUMERIC	Card payments
card_spend_amount	NUMERIC	Card volume
invitations_sent	NUMERIC	New users invited
settings_changed	NUMERIC	Config updates
support_tickets_raised	NUMERIC	Help requests
last_activity_time	TIMESTAMP	Most recent action
first_activity_time	TIMESTAMP	First action today
is_active_today	BOOLEAN	Any activity
has_transactions_today	BOOLEAN	Transaction flag
created_at	TIMESTAMP	Record creation

13. metrics_transactions_end_of_day

Purpose: Daily transaction and card usage aggregates

Grain: One row per day (system-wide)

Field Name	Data Type	Description
date	DATE	Metrics date
total_transaction_count	NUMERIC	All transactions
total_transaction_volume	NUMERIC	Total value
fps_count	NUMERIC	Faster payments
fps_volume	NUMERIC	FPS value
bacs_count	NUMERIC	BACS payments
bacs_volume	NUMERIC	BACS value
chaps_count	NUMERIC	CHAPS payments
chaps_volume	NUMERIC	CHAPS value
internal_transfer_count	NUMERIC	Between accounts
internal_transfer_volume	NUMERIC	Internal value
card_transaction_count	NUMERIC	Card payments
card_transaction_volume	NUMERIC	Card value
interest_payment_count	NUMERIC	Interest credits
interest_payment_volume	NUMERIC	Interest value
fee_transaction_count	NUMERIC	Fees charged
fee_transaction_volume	NUMERIC	Fee value
inbound_count	NUMERIC	All deposits
inbound_volume	NUMERIC	Deposit value
outbound_count	NUMERIC	All withdrawals
outbound_volume	NUMERIC	Withdrawal value
international_count	NUMERIC	Cross-border
international_volume	NUMERIC	International value
unique_accounts_transacting	NUMERIC	Active accounts
unique_businesses_transacting	NUMERIC	Active businesses
unique_persons_transacting	NUMERIC	Active users
failed_transaction_count	NUMERIC	Failed attempts
failed_transaction_volume	NUMERIC	Failed value
cancelled_transaction_count	NUMERIC	Cancelled
held_for_review_count	NUMERIC	Manual reviews
fraud_alerts_count	NUMERIC	Fraud flags
average_transaction_size	NUMERIC	Mean value
median_transaction_size	NUMERIC	Median value
largest_transaction	NUMERIC	Maximum value
card_merchant_categories	STRING	Top MCCs

Field Name	Data Type	Description
peak_hour	NUMERIC	Busiest hour
peak_hour_count	NUMERIC	Peak volume
created_at	TIMESTAMP	Record creation

14. mart_account_balances_end_of_day

Purpose: Point-in-time balance snapshots for all accounts

Grain: One row per account per snapshot time

Field Name	Data Type	Description
balance_at	TIMESTAMP	Snapshot timestamp (23:59:59 daily)
date	DATE	Balance date
account_id	STRING	Account identifier
business_id	STRING	For easier joins
companies_house_number	STRING	For easier joins
account_type	STRING	Spend/Earn
denomination	STRING	Currency (GBP)
settled_balance	NUMERIC	Settled amount
pending_incoming	NUMERIC	Unsettled credits
pending_outgoing	NUMERIC	Unsettled debits
available_balance	NUMERIC	Including overdraft
reserved_balance	NUMERIC	Held amounts
overdraft_used	NUMERIC	If overdrawn
accrued_interest	NUMERIC	For Earn accounts
accrued_fees	NUMERIC	Unpaid charges
is_overdrawn	BOOLEAN	Negative balance
is_dormant	BOOLEAN	No recent activity
created_at	TIMESTAMP	Record creation

15. mart_calendar

Purpose: Date dimension for consistent temporal analysis

Grain: One row per calendar day

Field Name	Data Type	Description
date	DATE	Calendar date
year_number	NUMERIC	Year (YYYY)
quarter_of_year	NUMERIC	Quarter (1-4)
month_of_year	NUMERIC	Month (1-12)
week_of_year	NUMERIC	Week (1-53)
day_of_year	NUMERIC	Day (1-366)
day_of_month	NUMERIC	Day (1-31)
day_of_week	NUMERIC	Day (1-7, Mon-Sun)
month_name	STRING	Full month name
month_name_short	STRING	3-letter month
day_of_week_name	STRING	Full day name
day_of_week_name_short	STRING	3-letter day
is_weekend	BOOLEAN	Saturday/Sunday
is_weekday	BOOLEAN	Monday-Friday
is_uk_bank_holiday	BOOLEAN	UK public holiday
uk_holiday_name	STRING	Holiday description
is_end_of_month	BOOLEAN	Last day of month
is_end_of_quarter	BOOLEAN	Last day of quarter
is_end_of_year	BOOLEAN	Last day of year
fiscal_year	NUMERIC	UK tax year
fiscal_quarter	NUMERIC	Tax quarter
relative_date	STRING	today/yesterday/etc

16. mart_sessions

Purpose: User authentication and activity sessions

Grain: One row per session

Field Name	Data Type	Description
session_id	STRING	Primary key - unique session
person_id	STRING	Person identifier
business_id	STRING	Business context
platform	STRING	web_app/mobile_app
device_type	STRING	desktop/mobile/tablet
device_os	STRING	Operating system
browser	STRING	If web
app_version	STRING	Application version
started_at	TIMESTAMP	Login time
ended_at	TIMESTAMP	Logout/timeout
duration_minutes	NUMERIC	Session length
authentication_type	STRING	hard/soft/silent
authentication_method	STRING	password/biometric/sso
ip_address	STRING	Origin IP
country	STRING	Geo-location
city	STRING	Geo-location
pages_viewed	NUMERIC	Screens visited
actions_performed	NUMERIC	Interactions
transactions_initiated	NUMERIC	Payments started
transactions_completed	NUMERIC	Payments finished
idle_time_minutes	NUMERIC	Inactive time
session_timeout_reason	STRING	If timed out
is_suspicious	BOOLEAN	Security flag
created_at	TIMESTAMP	Record creation

17. mart_interest_accruals

Purpose: Daily interest calculations (separate from transactions)

Grain: One row per accrual event

Field Name	Data Type	Description
accrual_id	STRING	Primary key - unique accrual
account_id	STRING	Earn account
business_id	STRING	For easier joins
account_type	STRING	Should be Earn Account
accrual_date	DATE	Calculation date
balance_used	NUMERIC	Balance for calculation
days_in_period	NUMERIC	Accrual period
annual_rate	NUMERIC	AER used
daily_rate	NUMERIC	Daily rate applied
gross_interest_rate	NUMERIC	Before compounding
amount_accrued	NUMERIC	Interest amount
cumulative_accrued	NUMERIC	Month-to-date
denomination	STRING	Currency (GBP)
is_leap_year_adjustment	BOOLEAN	366-day year
is_partial_period	BOOLEAN	Not full day
created_at	TIMESTAMP	Calculation time

18. mart_interest_credits

Purpose: Interest payments to customers (part of transactions but separated for clarity)

Grain: One row per interest payment

Field Name	Data Type	Description
transaction_id	STRING	Primary key - payment ID
account_id	STRING	Receiving account
business_id	STRING	For easier joins
account_type	STRING	Should be Earn Account
payment_date	DATE	Credit date
period_start_date	DATE	Interest period start
period_end_date	DATE	Interest period end
days_in_period	NUMERIC	Period length
average_balance	NUMERIC	Period average
interest_rate	NUMERIC	Rate applied
gross_amount	NUMERIC	Before tax
tax_withheld	NUMERIC	If applicable
net_amount	NUMERIC	Credited amount
denomination	STRING	Currency (GBP)
applied_at	TIMESTAMP	Payment time
inserted_at	TIMESTAMP	System time
is_internal_user	BOOLEAN	Test account
created_at	TIMESTAMP	Record creation

19. metrics_aggregates_end_of_day

Purpose: System-wide daily summary for cross-checking and high-level metrics

Grain: One row per day

Field Name	Data Type	Description
date	DATE	Metrics date
total_businesses	NUMERIC	All businesses
active_businesses	NUMERIC	With activity
new_businesses	NUMERIC	Joined today
total_persons	NUMERIC	All persons
active_persons	NUMERIC	Logged in today
new_persons	NUMERIC	Joined today
total_accounts	NUMERIC	All accounts
active_accounts	NUMERIC	With transactions
new_accounts	NUMERIC	Opened today
total_cards	NUMERIC	All cards
active_cards	NUMERIC	Used today
new_cards	NUMERIC	Activated today
total_balance_all_accounts	NUMERIC	System-wide
total_spend_balance	NUMERIC	Current accounts
total_earn_balance	NUMERIC	Savings accounts
net_balance_change	NUMERIC	Daily movement
total_transaction_count	NUMERIC	All transactions
total_transaction_volume	NUMERIC	Total value
total_deposits	NUMERIC	Inbound value
total_withdrawals	NUMERIC	Outbound value
unique_transacting_businesses	NUMERIC	Active businesses
unique_transacting_persons	NUMERIC	Active users
total_sessions	NUMERIC	All logins
total_session_minutes	NUMERIC	Engagement time
mobile_sessions_percentage	NUMERIC	Mobile share
total_invitations_sent	NUMERIC	New invites
total_invitations_accepted	NUMERIC	Accepted today
total_applications_started	NUMERIC	New applications
total_applications_submitted	NUMERIC	Completed apps
total_applications_approved	NUMERIC	Approved today
system_availability_percentage	NUMERIC	Uptime
api_response_time_ms	NUMERIC	Performance
created_at	TIMESTAMP	Record creation

20. mart_emails

Purpose: Email communications - both marketing and transactional

Grain: One row per email event

Field Name	Data Type	Description
iterable_id	STRING	Primary key - unique event ID
message_id	STRING	Email message ID
person_id	STRING	Recipient (segment_user_id)
email	STRING	Recipient email address
campaign_id	STRING	Campaign identifier
campaign_name	STRING	Campaign description
template_id	STRING	Email template
template_name	STRING	Template description
workflow_id	STRING	Automation workflow
workflow_name	STRING	Workflow description
channel_id	STRING	Sending channel
content_id	STRING	Content variant
email_subject	STRING	Subject line
email_service_provider	STRING	ESP used
sending_domain	STRING	From domain
event	STRING	Event type
event_text	STRING	Event description
created_at	TIMESTAMP	Event time
sent_at	TIMESTAMP	Send time
received_at	TIMESTAMP	Pipeline receipt
last_updated_at	TIMESTAMP	Last event update
has_email_opened	BOOLEAN	Open tracked
email_opened_at	TIMESTAMP	First open
has_email_clicked	BOOLEAN	Click tracked
email_clicked_at	TIMESTAMP	First click
clicks_count	NUMERIC	Total clicks
has_email_bounced	BOOLEAN	Delivery failed
bounce_type	STRING	hard/soft
bounced_message	STRING	Bounce reason
has_email_marked_as_spam	BOOLEAN	Spam complaint
marked_as_spam_at	TIMESTAMP	Complaint time
has_email_send_skipped	BOOLEAN	Not sent
skipped_cause	STRING	Skip reason
has_unsubscribed	BOOLEAN	Opted out
unsubscribed_at	TIMESTAMP	Opt-out time

Field Name	Data Type	Description
labels	STRING	Tag list
is_transactional	BOOLEAN	Not marketing
is_oaknorth_email	BOOLEAN	Internal recipient
email_data	STRING	JSON metadata
created_at_record	TIMESTAMP	Record creation

21. mart_push_alerts

Purpose: Mobile push notifications

Grain: One row per push event

Field Name	Data Type	Description
iterable_id	STRING	Primary key - unique event ID
message_id	STRING	Push message ID
person_id	STRING	Recipient (segment_user_id)
email	STRING	Associated email
device_id	STRING	Target device
platform_endpoint	STRING	Device token
campaign_id	STRING	Campaign identifier
campaign_name	STRING	Campaign description
template_id	STRING	Push template
template_name	STRING	Template description
workflow_id	STRING	Automation workflow
workflow_name	STRING	Workflow description
channel_id	STRING	Sending channel
content_id	STRING	Content variant
push_message	STRING	Notification text
payload_title	STRING	Notification title
payload_body	STRING	Notification body
event	STRING	Event type
event_text	STRING	Event description
created_at	TIMESTAMP	Event time
sent_at	TIMESTAMP	Send time
received_at	TIMESTAMP	Pipeline receipt
last_updated_at	TIMESTAMP	Last event update
has_push_opened	BOOLEAN	Open tracked
push_opened_at	TIMESTAMP	Open time
has_push_delivered	BOOLEAN	Delivered
push_delivered_at	TIMESTAMP	Delivery time
has_push_bounced	BOOLEAN	Failed
bounce_reason	STRING	Failure reason
is_payload_background_only	BOOLEAN	Silent push
message_type_id	STRING	Notification type
labels	STRING	Tag list
default_action_data	STRING	Action metadata
transactional_data	STRING	JSON metadata
is_oaknorth_email	BOOLEAN	Internal recipient

Field Name	Data Type	Description
created_at_record	TIMESTAMP	Record creation

Key Design Principles

Terminology Alignment

- **Person** not User (user_id mapped to person_id for consistency)
- **Business** not Company (except companies_house_number)
- **Removed/Closed** not Deleted for terminations
- **Platform** not banking_system for simplicity
- **End_of_day** not EOD for clarity

Data Type Standards

- **STRING:** All IDs, text fields, enums
- **NUMERIC:** All amounts, counts, scores
- **BOOLEAN:** All flags and binary states
- **TIMESTAMP:** Precise moments in time
- **DATE:** Calendar dates only

Naming Conventions

- **Tables:** Pluralized entities (businesses, persons, accounts)
- **IDs:** Explicit naming (person_id, business_id, never bare "id")
- **Timestamps:** _at suffix (created_at, activated_at)
- **Dates:** _on suffix for dates, _date for date fields
- **Booleans:** is_, has_, was_ prefixes
- **Counts:** _count suffix
- **Amounts:** _amount or _balance suffix

Relationship Management

- Persons relate to Businesses via bridge table
- Accounts belong to Businesses
- Cards belong to Persons and link to Accounts
- Clear parent-child relationships throughout

Optimization for AI

- Wide tables with pre-computed metrics

- Boolean flags for common filters
- Semantic field naming
- Minimal joins required for common queries
- Consistent patterns across similar entities